

Ablation Study for Number and Depth of Trees

Here we give a detailed overview of the sensitivity of VaSST w.r.t. number of trees K and depth D . We conducted an ablation study analyzing the impact of: (i) number of trees K and (ii) tree depth D . For each of the Feynman equations considered, VaSST is implemented for different values of

$$K \in \{2, 3, 4, 5\}, \quad D \in \{2, 3, 4, 5\},$$

keeping all other hyperparameters fixed. These new results will feature in the Appendix, along with a discussion of VaSST’s sensitivity with respect to K and D .

A highlight of the ablation study for two representative Feynman equations

$$P = \frac{\kappa A(T_2 - T_1)}{d}, \quad \Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right),$$

is as follows. Gaussian noise with standard deviation 0.1 has been added to the data. See Tables 1—4. The key findings are:

- **Performance stabilizes with model complexity.** For both equations, RMSE improves significantly when moving from $K = 2$ to $K \geq 3$, and stabilizes for $K \geq 4, D \geq 3$. This indicates that VaSST does not require large model sizes to achieve optimal predictive accuracy.
- **Low-capacity regimes can limit symbolic recovery.** As expected, insufficient model capacity leads to occasional failures in recovering the true expression. For

$$P = \frac{\kappa A(T_2 - T_1)}{d},$$

failure occurs at $(K = 2, D = 5)$, and for

$$\Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right),$$

failures occur at $(K = 2, D \in \{2, 4, 5\})$ and $(K = 3, D = 5)$. This highlights that both sufficient K and appropriate D are necessary for capturing compositional structure.

- **Runtime scales predictably with K and D .** Runtime increases approximately linearly in K and exponentially with D , reflecting the growth in the number of nodes. Importantly, moderate configurations (e.g., $K = 3, D = 3$) already achieve near-optimal RMSE, offering a favorable accuracy–efficiency trade-off.
- **Principled heuristics for choosing K and D .** Based on the ablation results, we include practical guidelines for selecting K and D . Since VaSST is designed to recover simple and interpretable scientific expressions, our recommendation follows an Occam’s razor principle: choose the smallest model capacity that attains stable RMSE and symbolic recovery. Empirically, $K \in \{3, 4\}$ and $D \in \{3, 4\}$ provide a favorable accuracy–interpretability–runtime trade-off, while larger depths increase runtime exponentially and may lead to unnecessarily complex symbolic expressions without substantial predictive gains.

Table 1: Runtime for $P = \frac{\kappa A(T_2 - T_1)}{d}$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	50.6	97.2	191.3	401.6
3	62.1	112.0	221.8	498.5
4	72.2	125.0	249.8	628.2
5	83.1	146.0	299.4	803.4

Table 2: RMSE for $P = \frac{\kappa A(T_2 - T_1)}{d}$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	0.1019	0.1002	0.1002	0.1070
3	0.1000	0.1011	0.1012	0.1019
4	0.1000	0.0999	0.1000	0.1000
5	0.0999	0.0999	0.0998	0.1007

Table 3: Runtime for $\Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	54.7	98.8	192.4	399.6
3	61.5	112.0	221.4	491.3
4	67.9	123.1	251.8	600.9
5	72.3	134.8	286.4	800.5

Table 4: RMSE for $\Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	0.2330	0.1012	0.1959	0.2349
3	0.1002	0.1019	0.1051	0.1918
4	0.1168	0.1002	0.1005	0.1012
5	0.1002	0.1005	0.1004	0.1010